- Ubuntu Server 22.04
- PostgreSQL 16
- Python3
- pgAdmin4
- Librerías Python
- Backup scripts
- SSL
- Exportaciones XML/JSON

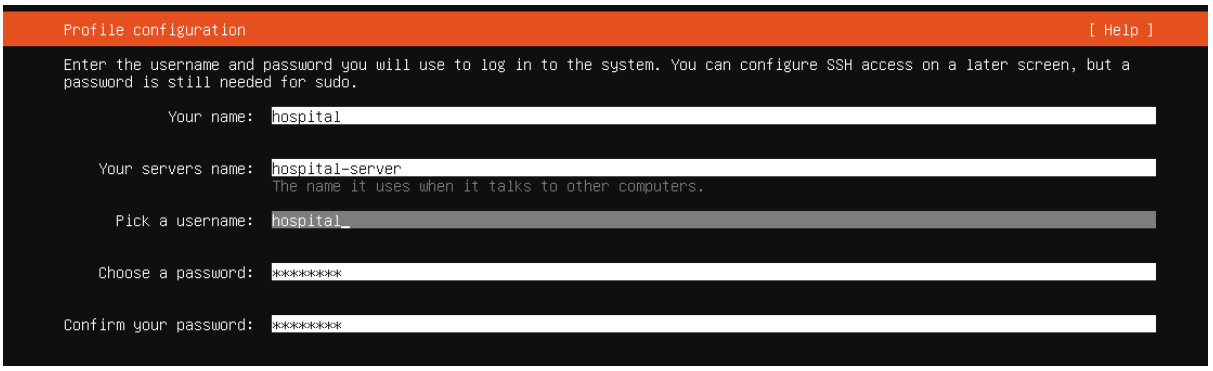
MV1 (Maquina virtual principal del projecte):
Nom MV: HospitalHarmoniaSV
Memoria: 4096 MB
Visio optica dinamica VIDI - 30 GB de RAM
2 **CPU's**
Adaptador 1: NAT
Adaptador 2: Adaptador pont

MV2 (Servidor replica)
Nom MV: HospitalReplica
- Alta disponibilidad
- Replicació
- Failover
- Backup real (.ova)

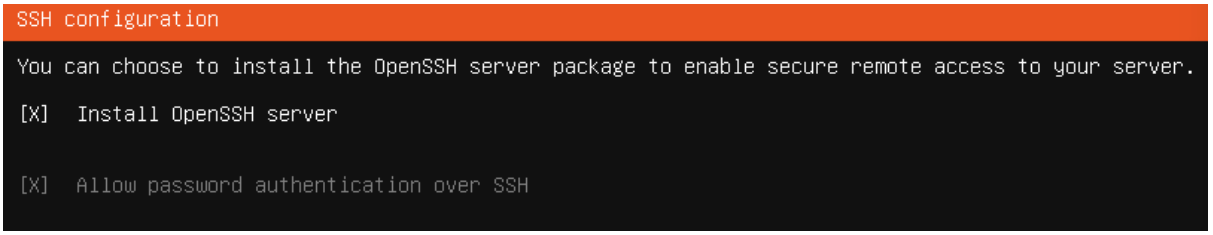
Power Bi: No va en la MV, va en el propi dispositiu Windows que se conecta a PostgreSQL de la MV.

Ara mateix crearem nomes HospitalHarmoniaSV amb Ubuntu + postgreSQL + Python perque más val 1 MV perfecta que dues trencades.

MV1 (Maquina virtual principal del projecte)



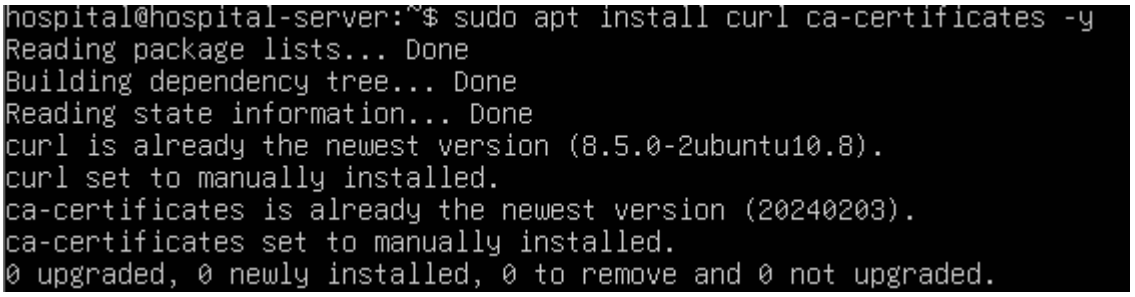
name: hospital
server name: hospital-server
username: hospital
password: P@ssw0rd



<u>Instal·lació PAS A PAS de PostgreSQL 16</u>

sudo apt update
sudo apt upgrade -y

Afegir el repositori oficial de PostgreSQL





sudo apt update

Instal·lar PostgreSQL 16



ip de connexio(xarxa dhcp de casa Gabriela): 192.168.1.24
ip de connexio clase: 192.168.56.112
navegador: http://192.168.1.24/
admin@admin.com
password: admin

Abans de fer la connexió al servidor en el postgres desde pgadmin-web
Modifiquem el següent nano:

```
  GNU nano 7.2                                        /etc/postgresql/16/main/postgresql.conf
#------------------------------------------------------------------------------
# CONNECTIONS AND AUTHENTICATION
#------------------------------------------------------------------------------

# - Connection Settings -

listen_addresses = '*'          # what IP address(es) to listen on;
```

I al nano /etc/postgresql/16/main/pg_hba.conf afegim la ultima linea:

```
host    all             all             192.168.1.0/24          md5
host    all             all             172.17.0.0/16           md5
```

Reincinem:

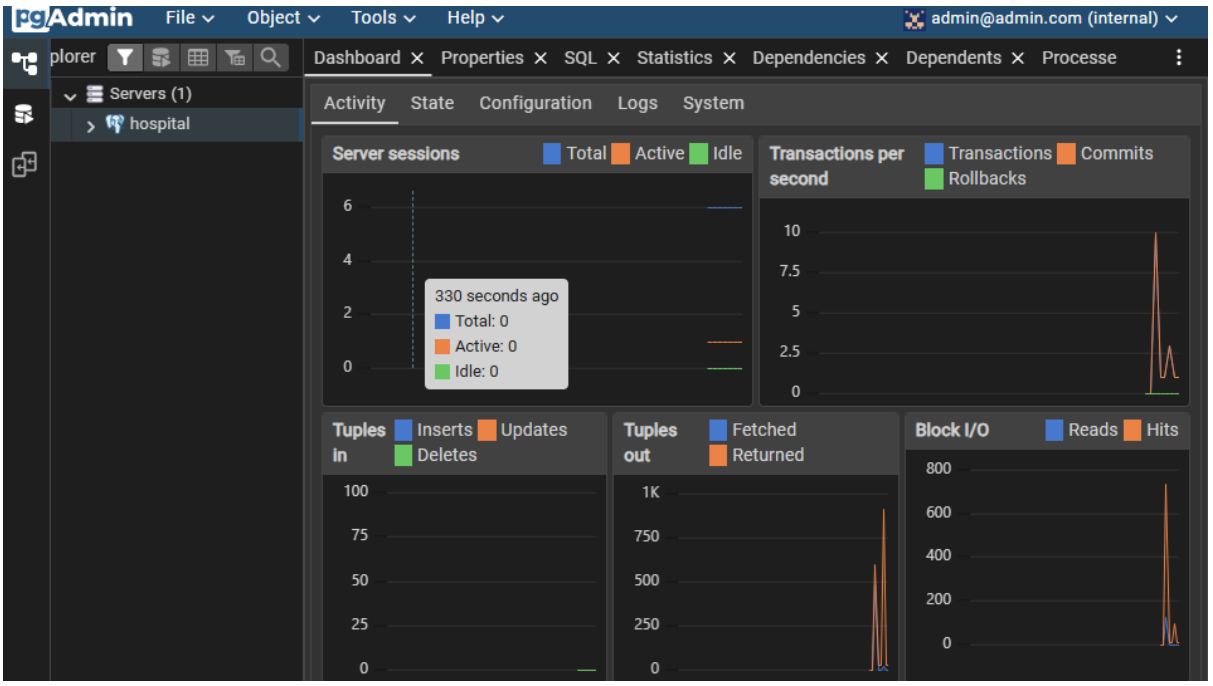
```
hospital@hospital-server:~$ sudo systemctl restart postgresql
hospital@hospital-server:~$
```

…

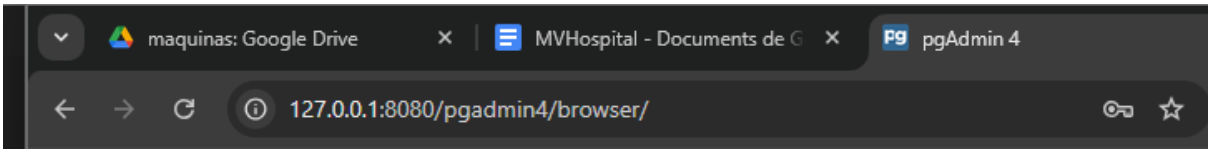
```
hospital@hospital-server:~$ sudo -u postgres psql
psql (16.13 (Ubuntu 16.13-0ubuntu0.24.04.1))
Type "help" for help.

postgres=# ALTER USER postgres WITH PASSWORD 'admin';
ALTER ROLE
postgres=#
```

Ara si farem la connexió:
Nom: hospital
Host: 192.168.1.24
Port: 5432
User: postgres
Password: admin (save password)
Database: postgres



PostgreSQL funcionant
pgAdmin funcionant
Connexió externa funcionant
sudo /usr/pgadmin4/bin/setup-web.sh: admin@admin.com P@ssw0rd
classe:



Ara configurar el postgresql amb ssl per que pgadmin, python o qualsevol client es conecti de forma segura.
Crear carpeta per certificats

```
hospital@hospital-server:~$ sudo mkdir /etc/postgresql/16/main/ssl
[sudo] password for hospital:
hospital@hospital-server:~$ sudo chmod 700 /etc/postgresql/16/main/ssl
hospital@hospital-server:~$
```

Generem clau privada i certificada

```
hospital@hospital-server:~$ sudo chown postgres:postgres /etc/postgresql/16/main/ssl
hospital@hospital-server:~$ sudo -u postgres bash'
> ^C
hospital@hospital-server:~$ sudo -u postgres bash
postgres@hospital-server:/home/hospital$ cd /etc/postgresql/16/main/ssl
postgres@hospital-server:/etc/postgresql/16/main/ssl$
```

Entrem a:

```
postgres@hospital-server:/etc/postgresql/16/main/ssl$ openssl genrsa -out server.key 2048
postgres@hospital-server:/etc/postgresql/16/main/ssl$ chmod 600 server.key
postgres@hospital-server:/etc/postgresql/16/main/ssl$ openssl req -new -x509 -days 365 -key server.key -out server.crt
```

Afegim:
Country name ES
State or Province catalunya
Location name Tordera

Organization Name Hospital
Organization Unit Name IT
Common Name 192.168.1.24
email address admin@admin.com

Comandes completes:

```
hospital@hospital-server:~$ sudo -u postgres bash
postgres@hospital-server:/home/hospital$ cd /etc/postgresql/16/main/ssl
postgres@hospital-server:/etc/postgresql/16/main/ssl$ openssl genrsa -out server.key 2048
cpostgres@hospital-server:/etc/postgresql/16/main/ssl$ chmod 600 server.key
postgres@hospital-server:/etc/postgresql/16/main/ssl$ openssl req -new -x509 -days 365 -key server.key -out server.crt
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:ES
State or Province Name (full name) [Some-State]:Catalunya
Locality Name (eg, city) []:Blanes
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Hospital Harmonia de Blanes
Organizational Unit Name (eg, section) []:IT
Common Name (e.g. server FQDN or YOUR name) []:192.168.1.24
Email Address []:admin@admin.com
```

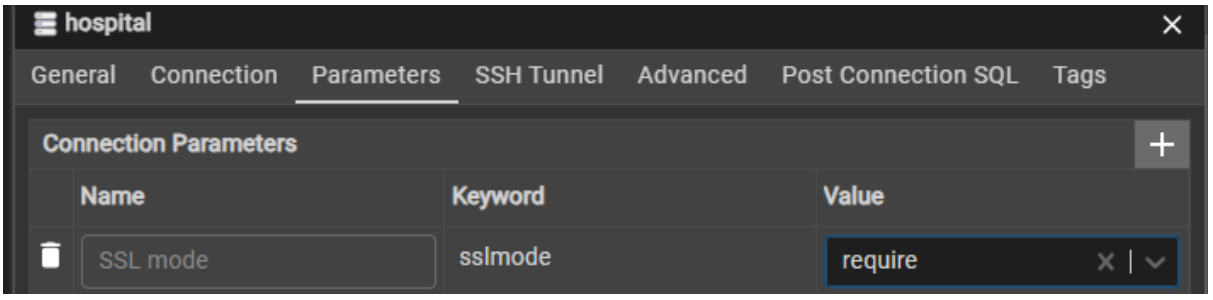
Activar SSL en postgresql.conf

```
ssl = on
#ssl_ca_file = ''
ssl_cert_file = 'ssl/server.crt'
#ssl_crl_file = ''
#ssl_crl_dir = ''
ssl_key_file = 'ssl/server.key'
```

Permitir conexions SSL a pg_hba.conf afeguin al final de l'arxiu les següents dos líneas:

```
hostssl all             all             192.168.1.0/24          md5
host    all             all             172.17.0.0/16           md5
```

Reiniciem el sistema postgresql i ajustem el protocol SSL en pgAdmin:



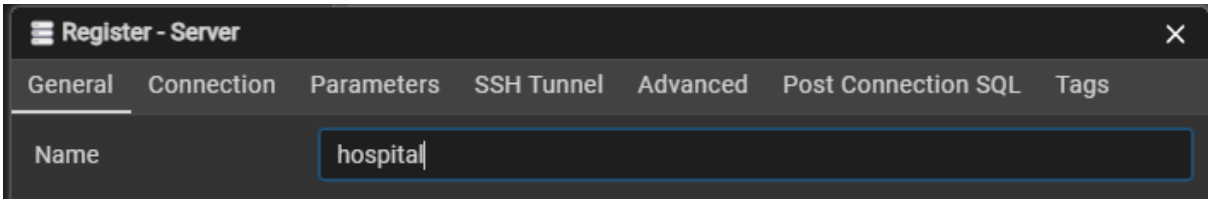
chown postgres:postgres → PostgreSQL solament pot llegir arxius que le perteneixen
chmod 700 → La carpeta SSL te que se solament excesible per l'usuari postgres
chmod 600 → Les claus i certificats tenen que ser llegibles solament per postgres.

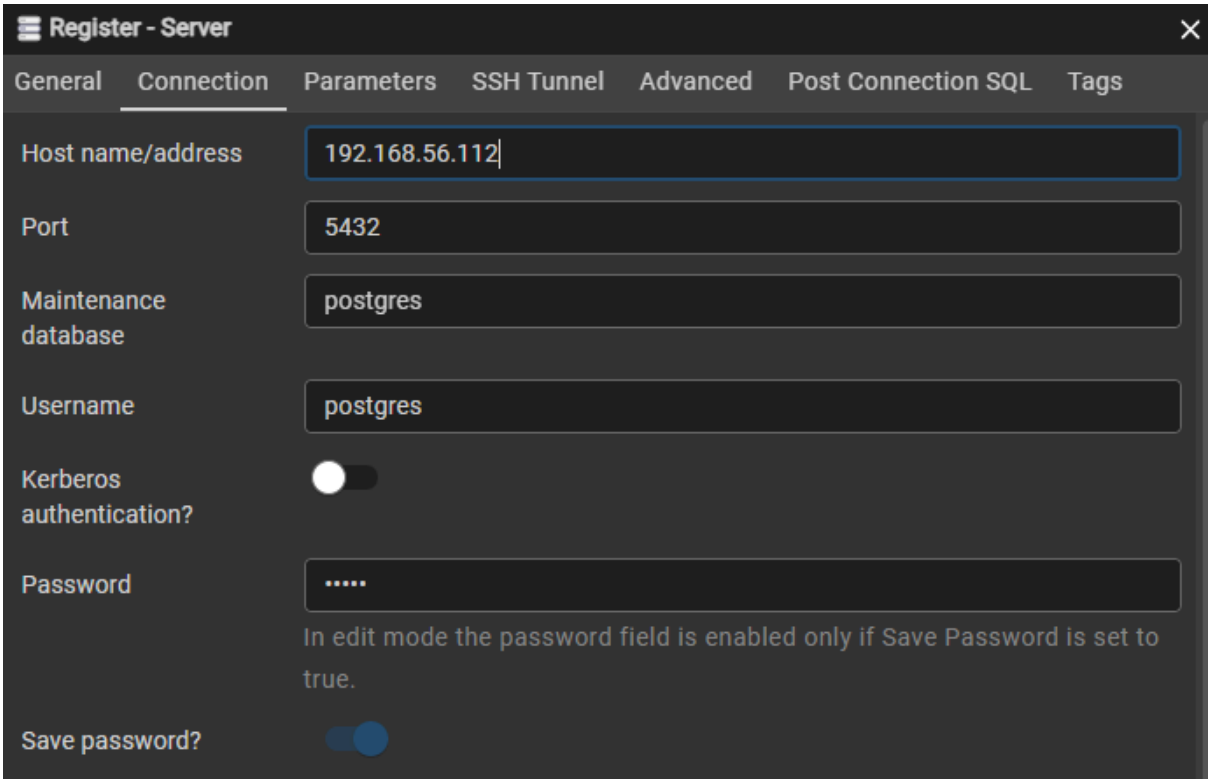
Si el grup o "altres" tenen permisos de lectura, postgreSQL es nega a arrancar per seguretat.

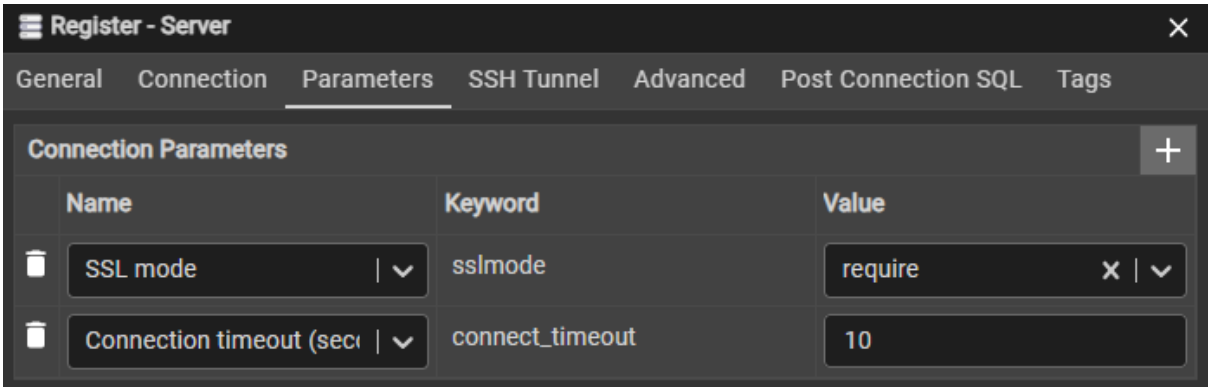
```
hospital@hospital-server:~$ sudo chown -R postgres:postgres /var/lib/postgresql/16/main/ssl
hospital@hospital-server:~$ sudo chmod 700 /var/lib/postgresql/16/main/ssl
hospital@hospital-server:~$ sudo chmod 600 /var/lib/postgresql/16/main/ssl/server.key
hospital@hospital-server:~$ sudo chmod 600 /var/lib/postgresql/16/main/ssl/server.crt
hospital@hospital-server:~$ sudo systemctl restart postgresql@16-main
hospital@hospital-server:~$ sudo systemctl status postgresql@16-main
● postgresql@16-main.service - PostgreSQL Cluster 16-main
     Loaded: loaded (/usr/lib/systemd/system/postgresql@.service; enabled-runtime; preset: enabled)
     Active: active (running) since Wed 2026-04-22 16:57:25 UTC; 16s ago
    Process: 4364 ExecStart=/usr/bin/pg_ctlcluster --skip-systemctl-redirect 16-main start (code=exited, status=0/SUCCESS)
   Main PID: 4369 (postgres)
      Tasks: 6 (limit: 4601)
     Memory: 18.4M (peak: 27.8M)
        CPU: 140ms
     CGroup: /system.slice/system-postgresql.slice/postgresql@16-main.service
             ├─4369 /usr/lib/postgresql/16/bin/postgres -D /var/lib/postgresql/16/main -c config_file=/etc/postgresql/16/main/postgresql.conf
             ├─4370 "postgres: 16/main: checkpointer "
             ├─4371 "postgres: 16/main: background_writer "
             ├─4373 "postgres: 16/main: walwriter "
             ├─4374 "postgres: 16/main: autovacuum launcher "
             └─4375 "postgres: 16/main: logical replication launcher "

Apr 22 16:57:23 hospital-server systemd[1]: Starting postgresql@16-main.service - PostgreSQL Cluster 16-main...
Apr 22 16:57:25 hospital-server systemd[1]: Started postgresql@16-main.service - PostgreSQL Cluster 16-main.
```

El servidor ja està trencat, estable i acceptant conexiones ssl.

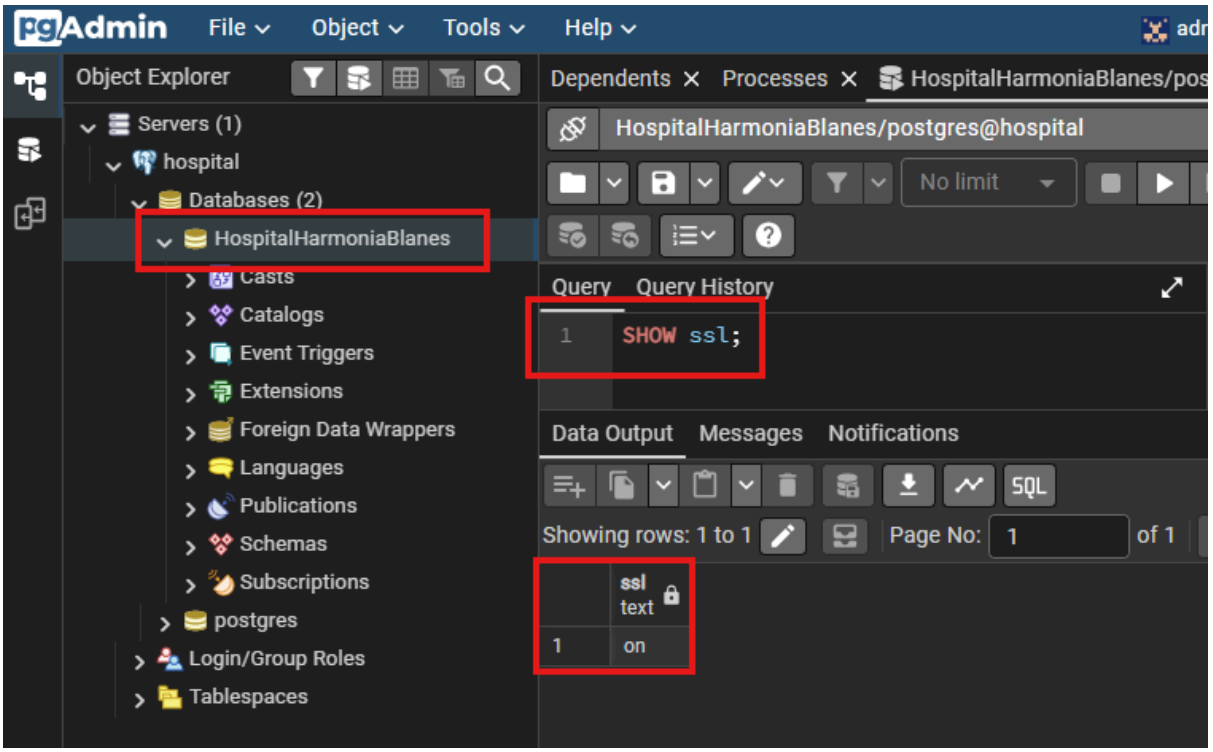






PostgreSQL está arrancan
SSL está funcionando
pgAdmin es conecta correctament
Ja podem treballar amb la base de dades sense problema

Verificació de que la sessió utilitza SSL:



Verificació de que la nostra sessió concreta esta xifrada:

